

Qatar-10b

R-0114

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-10_250601 · created 2026-07-16T04:23:49+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460827.795 +/- 0.012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.173 +/- 0.051
Transit depth (Rp/Rs)^2	2.99 +/- 1.76 [%]
Orbital Inclination (inc)	82.98 +/- 2.61
Airmass coefficient 1 (a1)	0.97 +/- 0.14
Airmass coefficient 2 (a2)	0.01 +/- 0.13
Scatter in the residuals of the lightcurve fit is	14.3 %
Best Comparison Star	#2 - [560, 408]
Optimal Aperture	2.45
Optimal Annulus	12.19
Transit Duration (day)	0.049 +/- 0.016

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.173 ± 0.051 vs 0.1265 (deviation 0.68σ)
Duration fit vs published	0.049 d vs 0.1154 d (deviation 3.1σ)
Mid-time O–C	22.5 min
Depth SNR	2.5
Residual scatter	14.3 %

Light curve

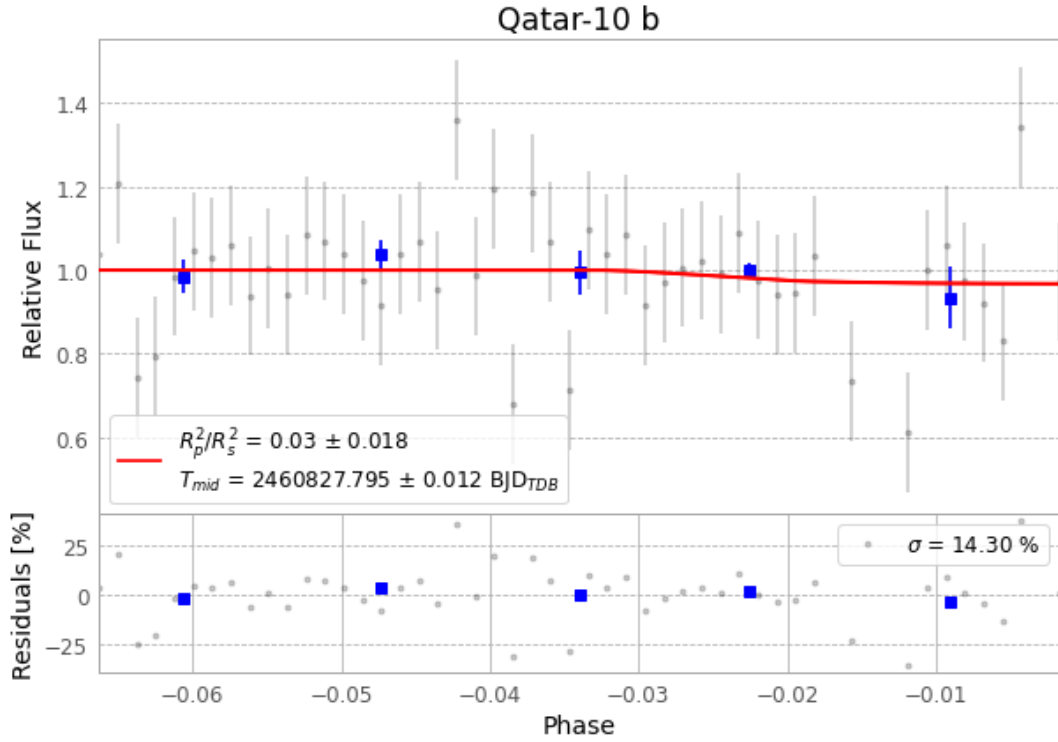


Figure 1 — FinalLightCurve_Qatar-10 b_2025-05-31.png (SHA-256 d3c91fed0aba9aa1...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [373, 179], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 70e1cef270975318...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit f0dcec7

Data provenance

52 FITS frames (34 MB), Qatar-10250601042715.FITS → Qatar-10250601070015.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-10-250601.log (970ef85ff16b...), FinalLightCurve_Qatar-10 b_2025-05-31.png (d3c91fed0aba...), FinalLightCurve_Qatar-10 b_2025-05-31.csv (0c99161f15ac...)

Compute resources

Wall time 145.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 04:23Z.